

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103
Regular & Supplementary Summer 2024**

Branch: Electronics and Telecommunication Engineering/ Electronics and Communication Engineering / Electronics Engineering

Course: B. Tech

Subject Code & Name: - BTETC402 / BTEXC402 Signals and Systems

Date:- 14/06/2024

Sem.:- IV

Marks: 60

Time:3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12

A) Two signals $x_1(t) = 10 \cos(100\pi t)$ and $x_2(t) = 10 \cos(50\pi t)$ are sampled at $f_s = 75$ Hz. Show that both sequences obtained after sampling are identical.	CO1	6
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B) 1. Calculate and draw even and odd parts of the signal: $x(t) = e^{-2t} \cos(t)$	CO1	6
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2. Function $x(t)$ is as shown in Fig. 1. Calculate and draw even and odd parts of the signal.

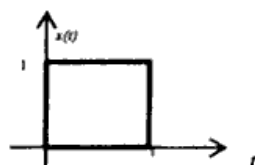


Fig. 1 : Given function $x(t)$

C) Function $x(t)$ is as shown in Fig. 2. Find a) $x(2t)$ b) $x(-t+2)$ c) $x(2t-2)$	CO1	6
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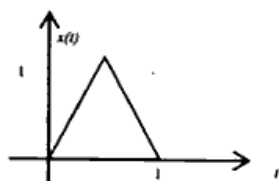


Fig. 2 : Given function $x(t)$

Q.2 Solve Any Two of the following.		12
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A) Derive the formula for convolution integral for a continuous time LTI system.	CO2	6
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B) Determine the following:	CO2	6
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1. $x(t) * u(t)$
2. $u(t) * u(t)$
3. $x(t) * \delta(t)$
4. $x(t) * \delta(t-3)$

C)	State and Explain relationship between the following LTI system properties and impulse response 1. Stability 2. Causality 3. Invertibility	CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Find the trigonometric Fourier Series for the following signal: $x(t) = 10t/2\pi$, $0 < t < 2\pi$	CO3	6
B)	If $x(t)$ is an even signal, derive the formulas for trigonometric Fourier series coefficients a_0 , a_n and b_n .	CO3	6
C)	Find the Fourier Transform of the signal: 1. $x(t) = \cos(\omega_0 t)$ 2. $x(t) = \sin(\omega_0 t)$	CO3	6
Q.4	Solve Any Two of the following.		12
A)	Consider a Causal LTI system with frequency response $H(w) = \frac{1}{3+jw}$ For a particular input $x(t)$, this system is observed to produce the output, $y(t) = e^{-3t}u(t) - e^{-4t}u(t)$. Determine $x(t)$.	CO3	6
B)	State and prove the following properties of DTFT: 1. Linearity 2. Time Shifting 3. Time Reversal 4. Frequency Shifting	CO4	6
C)	Determine the Inverse Z.T. of the following for $ z > 0.5$	CO4	6
	$X(z) = \frac{1}{1 - 0.5z^{-1}}$		12
Q. 5	Solve Any Two of the following.	CO4	6
A)	Derive the expression to obtain the Z-transform of the causal periodic sequence.	CO4	6
B)	Determine the Z.T. and sketch the ROC of the following signals: (a) $x(n) = (0.5)^{-n} u(n) + (2)^{-n} u(n)$ (b) $x(n) = -(0.5)^{-n} u(-n-1) + (2)^{-n} u(-n-1)$	CO4	6
C)	For the following system functions, find $h(n)$ and check whether the corresponding system is causal, anticausal, or noncausal. 1. $H(s) = \frac{1}{s^2+5s+6}$ $R\{s\} > -2$ 2. $H(s) = \frac{1}{s^2+5s+6}$ $R\{s\} < -3$ 3. $H(s) = \frac{1}{s^2+5s+6}$ $-2 < R\{s\} < -3$		

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